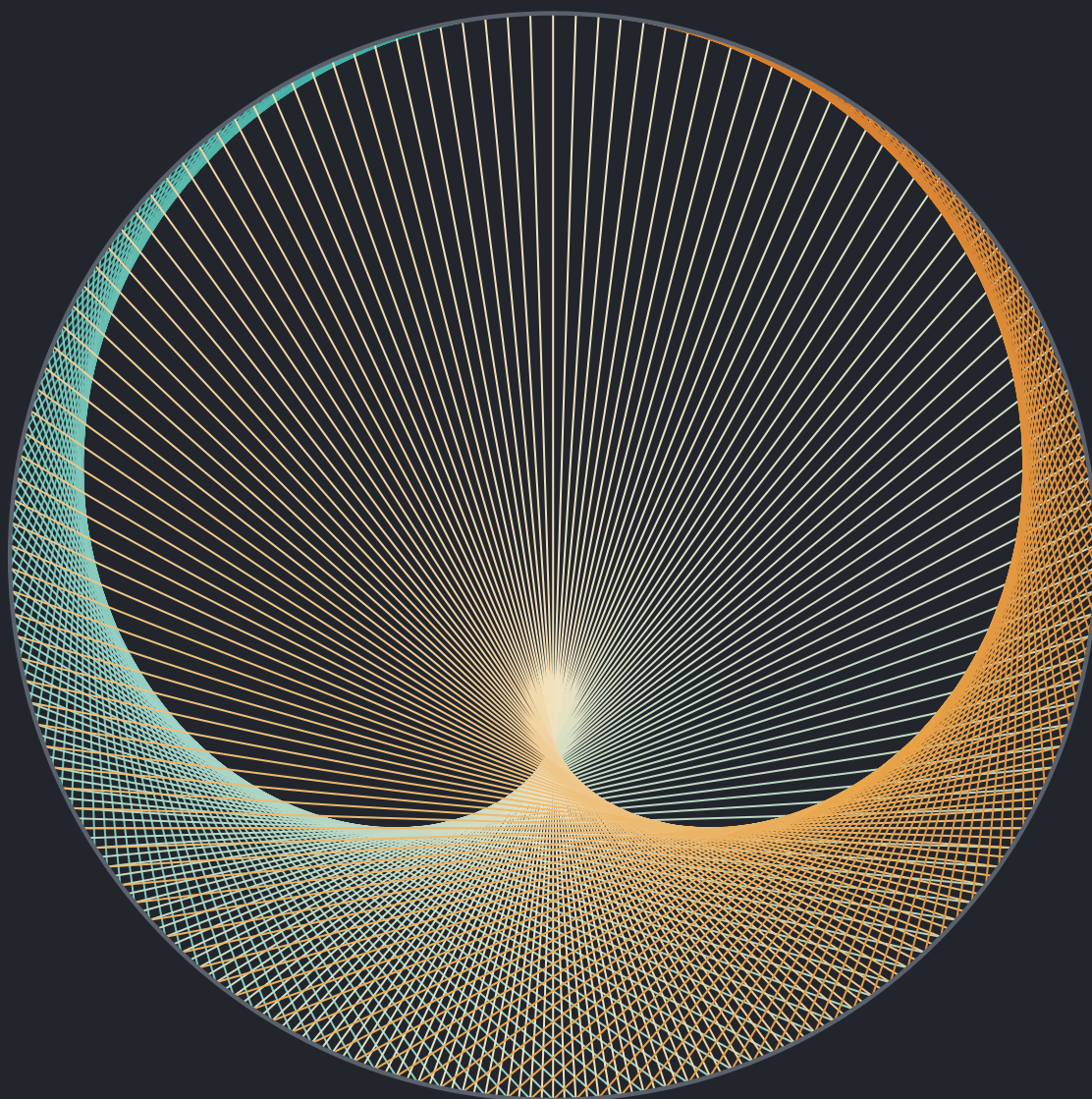


YEAR 7 MATHEMATICS

Chapter 1

Integers

LEARN IT / PRACTISE IT / REMEMBER IT



1.01



1.02



1.03



1.04



1.05



1.06



1.07



1.08



1.09



1.10



Review

On the cover: the 2 times table drawn on a circle of 300 points. Each lesson has its own times-table circle.

Every lesson is **one double-sided sheet**. Each new exercise starts with a dark number block and its own times-table circle. **Page 1: learn it. Page 2: practise it.**

EXAMPLE

Shaded. Your teacher works it out with you. **Copy their working** onto the lines.

**YOUR TURN**

White, with a strong border. **Now you try** one just like it, on your own.

- Do now** Start straight away, from memory.
- Think first** Have a go before the lesson. An example gives the answer.
- Words and Big idea** Fill in the Big idea with your teacher.
- Explain** Tick the better reason. Be ready to say why.
- Check** Everyone answers, so your teacher knows who is ready.
- Spot the pattern** One thing changes each time. Watch the answers.
- Mixed practice** Mostly today's skill. Check with the mixed-up answers.
- Go further** Finished? Questions that make you think harder.
- Exit ticket** Three questions on your own. Then your marked score.

PAGE 1: LEARN IT

Exercise number	
Do now	
Think first	
Words · Big idea	
Example	Your turn
Check	

PAGE 2: PRACTISE IT

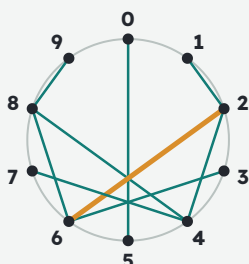
Lesson
Spot the pattern
Mixed practice
Go further
Exit ticket

Contents and progress tracker

	LESSON	PAGE	DATE	DONE
	1.01 Integers and opposites	3		☐
	1.02 The number line	5		☐
	1.03 Comparing and ordering integers	7		☐
	1.04 Adding integers	9		☐
	1.05 Subtracting integers	11		☐
	1.06 Multiplying integers	13		☐
	1.07 Dividing integers	15		☐
	1.08 Order of operations with integers	17		☐
	1.09 Integers on a calculator	19		☐
	1.10 Integer problems	21		☐
	R Chapter review	23		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

**About the circles**

Put 10 points around a circle, numbered 0 to 9. Join each point to the point **twice** its number. After 9, count on from 0 again: so 6 joins to 12, which is point 2. Use 300 points and the lines draw the heart shape on the cover. Lesson 1 uses the 2 times table, lesson 2 the 3 times table, and so on. Count the points of each shape: always one less than the times table.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Integers and opposites

Today I will use integers and find opposites.

NAME _____

CLASS _____

DATE _____

1.01

DO NOW From memory. No notes.

1 Order smallest to largest: 12, 3, 30, 0

2 What is $8 - 8$?3 It is 0°C . It gets 4 degrees colder. What is the temperature?

4 What number is halfway between 0 and 10?

from Year 6

from Year 6

from Year 6

from Year 6

THINK FIRST Have a go.

A lift is at the ground floor, level 0. It goes down 2 floors to the car park. Which level is that?

Check it in Example 1.

WORDS

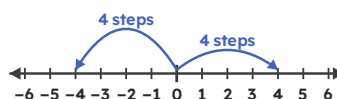
integer: a whole number: ...

 $-2, -1, 0, 1, 2, \dots$

opposite: same distance from 0, other side.

 $-(-5)$: say "the opposite of negative five".**BIG IDEA** Fill in the gaps with your teacher.

Negative numbers are to the _____ of 0 on a number line. A number and its _____ are the same _____ from 0, but on the other side.

**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

Write an integer for 5 m below sea level.

→

YOUR TURN 1

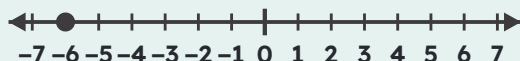
Write an integer for a loss of \$8.

a loss: _____

= _____

EXPLAIN

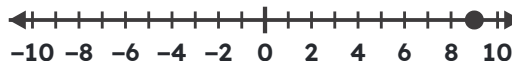
Why is 5 m below sea level a negative number?

☐ A Sea level is 0, and below 0 is negative.☐ B Going down means taking away.**EXAMPLE 2**The opposite of -6 

→

YOUR TURN 2

The opposite of 9



___ is ___ steps right of ___

EXPLAINHow does the number line show that 6 is the opposite of -6 ?☐ A 6 is bigger than -6 .☐ B Both are 6 steps from 0, on different sides.**EXAMPLE 3** $-(-7)$

→

YOUR TURN 3 $-(-11)$ **EXPLAIN**Why is $-(-7)$ equal to 7?☐ A Two negatives always make a positive.☐ B It means the opposite of -7 , and that is 7.**CHECK** Everyone answers. Circle one letter, then hold it up.What is the opposite of -8 ?(A) -8

(B) 8

(C) 0

(D) -7

**SPOT THE PATTERN** Find each one. Each question changes one thing.

- 1 opposite of 3 = _____ 2 opposite of 2 = _____ 3 opposite of 1 = _____ 4 opposite of 0 = _____
- 5 opposite of -1 = _____ 6 opposite of -2 = _____ 7 $-(-2) =$ _____ 8 $-(-20) =$ _____

What do you notice? The number goes down by 1 each time. What happens to its opposite?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 What is the opposite of 7? 2 What is the opposite of -3? 3 What is the opposite of 12?

4 Count back by 2s: 6, 4, 2, 0, ...
What are the next two numbers?

5 What is the opposite of -25?

6 $-(-4)$

from Year 6

7 $-(-15)$

8 What is the opposite of 0?

9 It is 2°C . It gets 5 degrees colder. What is the temperature now?

from Year 6

10 $-(-1)$

11 Write an integer for 6 m below sea level.

12 Write an integer for a gain of \$10.

13 What number is halfway between 0 and 20?

14 Which is further from 0: -7 or 5?

15 Which two integers are 10 steps from 0?

from Year 6

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.

15 25 -12 -7 -7 10 1 0 4 -2, -4 3 10 and -10 10 -6 -3°C

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Mia says "the opposite of a number is always negative". Is she right? Explain.

- 2 Two opposites are 18 steps apart. What are they?

- 3 Which integers from -5 to 5 are more than 3 steps from 0?

**EXIT TICKET** On your own. Hand it in.

- 1 Write an integer for 4°C below zero. 2 What is the opposite of -13? 3 $-(-2)$

Today I can use integers and find opposites. **After marking:** _____ / 3

The number line

Today I will read number lines and find distances.

NAME _____

CLASS _____

DATE _____

1.02

DO NOW From memory. No notes.

1 What is the opposite of -4?

2 $-(-10)$

3 Write an integer for 3 floors below ground.

4 Count by 5s: 15, 20, 25, ... What comes next?

from 1.01

from 1.01

from 1.01

from Year 6

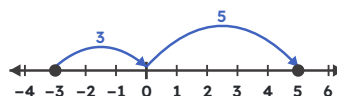
THINK FIRST Have a go.

A dot is halfway between -4 and 0. What number is it at?

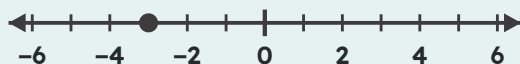
Check it in Example 3.

WORDS**scale:** how much each gap is worth.**distance:** how many steps apart. Never negative.**halfway:** the same distance from both ends.**BIG IDEA** Fill in the gaps with your teacher.

Numbers get _____ to the right. To find a distance, count the _____, not the marks. If the trip passes 0, count to _____, then count _____.

**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

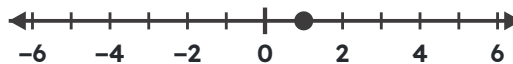
Which number is at the dot?



→

YOUR TURN 1

Which number is at the dot?

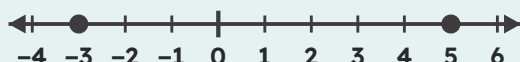


one gap right of ____

= ____

EXPLAIN Not every mark has a number. How can you find the number at the dot?☐ A Each gap is 1, so count gaps from a label.☐ B Count the marks from the left end.**EXAMPLE 2**

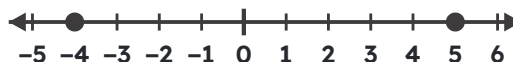
How far is it from -3 to 5?



→

YOUR TURN 2

How far is it from -4 to 5?



____ to ____, ____ from ____

EXPLAIN Why do we add the 3 and the 5?☐ A We always add the two numbers we are given.☐ B It is 3 steps to 0, then 5 more from 0.**EXAMPLE 3**

Which integer is halfway between -8 and 2?

→

YOUR TURN 3

Which integer is halfway between -6 and 4?

EXPLAIN How can you check that -3 is halfway between -8 and 2?☐ A -3 is 5 steps from -8 and 5 steps from 2.☐ B -3 is between -8 and 2.**CHECK** Everyone answers. Circle one letter, then hold it up.

How far apart are -4 and 3?

☐ (A) 1
 ☐ (B) 7
 ☐ (C) -7
 ☐ (D) -1

**SPOT THE PATTERN**

How far apart are the two numbers? Each question moves one end.

- 1 -1 to $3 =$ _____ 2 -2 to $3 =$ _____ 3 -3 to $3 =$ _____ 4 -4 to $3 =$ _____
- 5 -4 to $2 =$ _____ 6 -4 to $1 =$ _____ 7 -4 to $0 =$ _____ 8 -4 to $-1 =$ _____

What do you notice? In 1 to 4, one end moves 1 step further out. In 5 to 8, it moves 1 step back in. What happens to the distance?

**MIXED PRACTICE**

Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 How far is it from -2 to 3 ? 2 How far is it from -5 to 1 ? 3 How far is it from -4 to -1 ?
- 4 What is the opposite of -9 ? 5 Start at -3 . Move 5 right. Where do you land? 6 Start at 4 . Move 9 left. Where do you land?
- 7 Start at -6 . Move 2 left. Where do you land? 8 Which integer is halfway between -4 and 2 ? 9 $-(-3)$
- 10 Which integer is halfway between -9 and -1 ? 11 A number line goes $-20, -10, 0, \dots$ What are the next two numbers? 12 Point A is at -6 . Point B is 10 to the right of A. Where is B?
- 13 Write an integer for a loss of \$7. 14 A lift goes from level -3 to level 6. How many floors is that? 15 How far is it from -8 to 7 ?

from 1.01

from 1.01

from 1.01

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.

-8 2 3 5 9 floors 4 -5 -1 -5 9 6 15 -7 $10, 20$ 3

**GO FURTHER**

Finished? Try these. Explain your thinking.

- 1 Two integers are 12 apart. Halfway between them is -2 . What are they?
- 2 Start at -5 . Move 3 right, then 8 left, then 6 right. Where do you land? How many steps did you walk?
- 3 How many integers are between -6 and 4 , not counting -6 and 4 ?

**EXIT TICKET**

On your own. Hand it in.

- 1 How far is it from -7 to 2 ? 2 Which integer is halfway between -10 and 4 ? 3 Start at 3 and move 8 left. Where do you land?

Today I can read number lines and find distances. After marking: _____ / 3

Comparing and ordering integers

Today I will compare and order integers.

NAME _____

CLASS _____

DATE _____

1.03

DO NOW From memory. No notes.**1** How far is it from -5 to 3 ?

from 1.02

2 Start at -1 and move 6 left. Where do you land?

from 1.02

3 What is the opposite of -15 ?

from 1.01

4 Which integer is halfway between -6 and 2 ?

from 1.02

THINK FIRST Have a go.Which is colder: -8°C or -3°C ? So which number is smaller?

Check it in Example 1.

WORDS

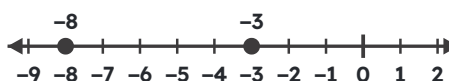
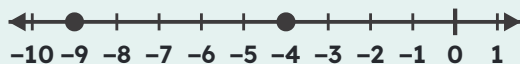
< means "is less than".

> means "is greater than".

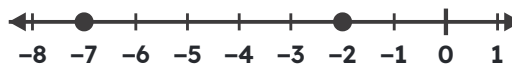
ascending: smallest to largest.

BIG IDEA Fill in the gaps with your teacher.

On a number line, the number further _____ is larger. So -3 is _____ than -8 , even though 8 is more than 3 . Colder temperatures are _____ numbers.

**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**Which is larger: -9 or -4 ?

→

YOUR TURN 1Which is larger: -2 or -7 ?

___ is right of ___

larger: ___

EXPLAIN Sam says -9 is larger than -4 , because 9 is more than 4 . Is Sam right?☐ **A** Yes, because 9 is more than 4 .☐ **B** No: -9 is further left, so it is smaller.**EXAMPLE 2**Write < or >: -6 ☐ -1

→

YOUR TURN 2Write < or >: 0 ☐ -5

___ is right of ___

EXPLAIN How did you choose between < and >?☐ **A** The open side faces the larger number, -1 .☐ **B** It points at the number with the bigger digit.**EXAMPLE 3**Order smallest to largest: $3, -5, 0, -2$

→

YOUR TURN 3Order smallest to largest: $-1, 4, -6, 2$ **EXPLAIN** Why does -5 come before -2 in the list?☐ **A** -5 is further left than -2 , so it is smaller.☐ **B** 5 is more than 2 .**CHECK** Everyone answers. Circle one letter, then hold it up.

Which is the smallest?

☐ **A** -2 ☐ **B** 0 ☐ **C** -9 ☐ **D** -5

**SPOT THE PATTERN** Write $<$ or $>$. Each question changes one thing.

- 1 $3 \square 5$ _____ 2 $-3 \square 5$ _____ 3 $-3 \square -5$ _____ 4 $3 \square -5$ _____
- 5 $-5 \square -3$ _____ 6 $-30 \square -5$ _____ 7 $-5 \square 0$ _____ 8 $0 \square -30$ _____

What do you notice? Look at 1 and 3. Both numbers became negative. What happened to the symbol?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 Write $<$ or $>$: $-4 \square 2$ 2 Write $<$ or $>$: $-6 \square -1$ 3 Write $<$ or $>$: $0 \square -3$

- 4 How far is it from -11 to -2 ? 5 Write $<$ or $>$: $-8 \square -9$ 6 Write $<$ or $>$: $-20 \square -19$

from 1.02

- 7 Which is smaller: -7 or -3 ? 8 Which is larger: -15 or -51 ? 9 What is the opposite of -14 ?

from 1.01

- 10 Order smallest to largest:
 $3, -2, -5$ 11 Order smallest to largest:
 $-1, -8, 0, -4$ 12 Which is colder: -15°C
or -12°C ?

- 13 Which integer is halfway
between -10 and 2 ? 14 Write an integer that is
between -5 and -3 . 15 Order largest to smallest:
 $-4, 7, -10, 0$

from 1.02

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order. $-7 > > < -4 -15^{\circ}\text{C} -5, -2, 3 -15 < 9 < 7, 0, -4, -10 -4 -8, -4, -1, 0 14$ **GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Always, sometimes or never? "A negative number is smaller than a positive number." Explain.

- 2 I am an odd integer. I am less than -2 and greater than -6 . What could I be?

- 3 Order these coldest first: $-7^{\circ}\text{C}, 4^{\circ}\text{C}, -12^{\circ}\text{C}, 0^{\circ}\text{C}$. How far apart are the coldest and warmest?

**EXIT TICKET** On your own. Hand it in.

- 1 Write $<$ or $>$: $-8 \square -3$ 2 Order smallest to largest:
 $2, -9, -1, -4$ 3 Which is warmer: -6°C
or -10°C ?

Today I can compare and order integers. After marking: _____ / 3

Adding integers

Today I will add positive and negative integers.

NAME _____

CLASS _____

DATE _____

1.04

DO NOW From memory. No notes.1 $-7 \square -2$ Write $<$ or $>$.

2 Order smallest to largest: 4, -6, 0, -1

3 What is the opposite of -9?

4 Start at -4. Move 6 right. Where do you land?

from 1.03

from 1.03

from 1.01

from 1.02

THINK FIRST Have a go.It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon?

Check it in Example 1.

WORDS

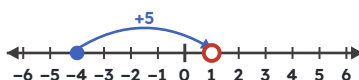
sum: the answer to an addition.

-3: say "negative three".

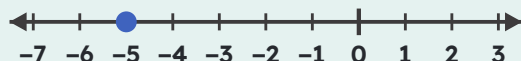
zero pair: +1 and -1 make 0.

BIG IDEA Fill in the gaps with your teacher.

Start at the _____ number. Adding a positive: move _____. Adding a negative: move _____, which is the same as _____.

**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

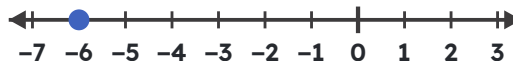
$-5 + 3$



→

YOUR TURN 1

$-6 + 4$

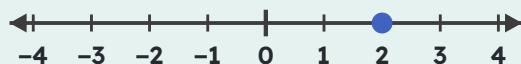


start at ____, move ____ right

= ____

EXPLAIN Why do we move right for $-5 + 3$?☐ A We are adding a positive number.☐ B The answer must be positive.**EXAMPLE 2**

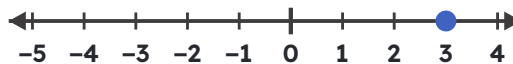
$2 + (-5)$



→

YOUR TURN 2

$3 + (-7)$



start at ____, move ____ left

EXPLAIN Why do $2 + (-5)$ and $2 - 5$ give the same answer?☐ A Adding always makes a number bigger.☐ B Both start at 2 and move 5 left.**EXAMPLE 3**

$-4 + (-3)$

→

YOUR TURN 3

$-2 + (-6)$

EXPLAIN Tom says $-4 + (-3) = 7$. What has Tom done wrong?☐ A He used "two negatives make a positive".☐ B He should have worked out $4 - 3$.**CHECK** Everyone answers. Circle one letter, then hold it up.Work out $-2 + 5$

(A) -7

(B) 3

(C) -3

(D) 7

**SPOT THE PATTERN** Each question changes one thing. Work them out, then look down the answers.

- 1 $-3 + 3 =$ _____ 2 $-3 + 2 =$ _____ 3 $-3 + 1 =$ _____ 4 $-3 + 0 =$ _____
- 5 $-3 + (-1) =$ _____ 6 $-3 + (-2) =$ _____ 7 $-3 + (-3) =$ _____ 8 $-30 + (-30) =$ _____

What do you notice? The number being added goes down by 1 each time. What happens to the answer?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 $-5 + 2$ 2 $-3 + 7$ 3 $-8 + 8$
- 4 Write $<$ or $>$: $-8 \square -5$ 5 $4 + (-6)$ 6 $2 + (-9)$
- 7 $-3 + (-4)$ 8 $-7 + (-2)$ 9 How far apart are -8 and 5 ?
- 10 $15 + (-20)$ 11 $-12 + 5$ 12 A diver is at -12 m. She rises 5 m. Where is she now?
- 13 What is the opposite of -12 ? 14 $-6 + \square = -1$ 15 It is -9°C . It rises 4°C , then 7°C more. What is the temperature now?

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order. -7 -2 0 -3 5 -7 m -5 -9 -7 $<$ 4 2°C 12 -7 13 **GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Always, sometimes or never true? "Adding a negative number makes the answer smaller." Explain.

- 2 Find two integers that add to
- -3
- . Then find two more, where one of them is positive.

- 3 Start at
- -4
- . Add 2, 6 and
- -5
- in any order. Where do you finish? Does the order matter?

**EXIT TICKET** On your own. Hand it in.

- 1 $-7 + 3$ 2 $4 + (-9)$ 3 It is -6°C . It warms up 10°C . What is the temperature now?

Today I can add positive and negative integers. After marking: _____ / 3

Subtracting integers

Today I will subtract positive and negative integers.

NAME _____

CLASS _____

DATE _____

1.05

DO NOW From memory. No notes.

1 $-7 + 4$

2 $5 + (-8)$

3 Write $<$ or $>$: $-6 \square -2$

4 What is the opposite of -5 ?

from 1.04

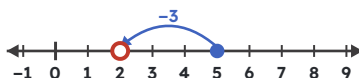
from 1.04

from 1.03

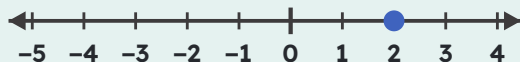
from 1.01

THINK FIRST Have a go.It is 3°C . It gets 5 degrees colder. What is the temperature now?

Check it in Example 1.

WORDS**difference:** the answer to a subtraction. $5 - (-3)$: say "five subtract negative three".**opposite:** swap the sign: 3 and -3 .**BIG IDEA** Fill in the gaps with your teacher.Subtracting is the same as adding the _____. $5 - 3 = 5 + (-3)$: move _____. $5 - (-3) = 5 + 3$: move _____.**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

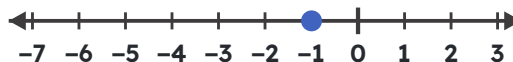
$2 - 6$



→

YOUR TURN 1

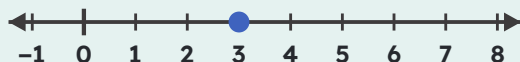
$-1 - 4$



start at ____, move ____ left

 $=$ ____**EXPLAIN** Why is the answer to $2 - 6$ negative?☐ A Subtracting always gives a negative answer.☐ B We take away more than we have, so we pass 0.**EXAMPLE 2**

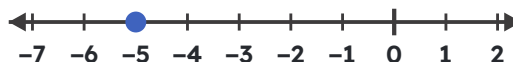
$3 - (-4)$



→

YOUR TURN 2

$-5 - (-2)$

 $=$ ____ + ____**EXPLAIN** Why can we rewrite $3 - (-4)$ as $3 + 4$?☐ A Subtracting -4 is adding its opposite, 4.☐ B Two minus signs side by side cancel out.**EXAMPLE 3**

$-6 - (-9)$

→

YOUR TURN 3

$-8 - (-3)$

EXPLAIN Ali says $3 - (-2) = 1$. What has Ali done wrong?☐ A Ali should have started at -2 .☐ B Ali took away 2 instead of -2 .**CHECK** Everyone answers. Circle one letter, then hold it up.Work out $4 - (-3)$
☐ (A) 1
 ☐ (B) 7
 ☐ (C) -7
☐ (D) -1

**SPOT THE PATTERN** Each question subtracts 1 less than the one before.

$$1 \quad 5 - 3 = \underline{\hspace{2cm}} \quad 2 \quad 5 - 2 = \underline{\hspace{2cm}} \quad 3 \quad 5 - 1 = \underline{\hspace{2cm}} \quad 4 \quad 5 - 0 = \underline{\hspace{2cm}}$$

$$5 \quad 5 - (-1) = \underline{\hspace{2cm}} \quad 6 \quad 5 - (-2) = \underline{\hspace{2cm}} \quad 7 \quad 5 - (-3) = \underline{\hspace{2cm}} \quad 8 \quad -5 - (-3) = \underline{\hspace{2cm}}$$

What do you notice? The number being subtracted goes down by 1 each time. What happens to the answer?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

1 $3 - 7$

2 $0 - 5$

3 $-2 - 4$

4 $-3 + (-6)$

5 $5 - (-2)$

6 $-3 - (-8)$

from 1.04

7 $-4 - (-4)$

8 $-9 - (-2)$

9 Write $<$ or $>$: $-12 \square -8$

from 1.03

10 $-12 - 8$

11 $7 - (-7)$

12 It is -4°C . It gets 7°C colder. What is the temperature now?

13 $-6 + 10$

14 How much warmer is 6°C than -2°C ?

15 $10 - (-5) - 20$

from 1.04

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.0 7 -6 -4 8°C -11°C -20 -7 5 -9 -5 -5 4 14 <**GO FURTHER** Finished? Try these. Explain your thinking.

1 Is $7 - (-2)$ the same as $7 + 2$? Use a number line to explain.

2 Find a positive integer and a second integer whose difference (first - second) is -5.

3 A submarine is at -120 m. A plane is at 350 m. How far apart are they?

**EXIT TICKET** On your own. Hand it in.

1 $-2 - 7$

2 $6 - (-5)$

3 $-4 - (-9)$

Today I can subtract positive and negative integers. After marking: $\underline{\hspace{2cm}}$ / 3

Multiplying integers

Today I will multiply integers and choose the sign.

NAME _____

CLASS _____

DATE _____

1.06

DO NOW From memory. No notes.

1 $-3 - 5$

2 $2 - (-6)$

3 $(-4) + (-4) + (-4)$

4 7×8

from 1.05

from 1.05

from 1.04

from Year 6

THINK FIRST Have a go.

You lose \$3 a day for 4 days. By how much has your money changed?

Check it in Example 1.

WORDS

product: the answer to a multiplication.

same signs: both + or both -.

different signs: one + and one -.

BIG IDEA Fill in the gaps with your teacher.

Multiply the numbers, then choose the sign. _____ signs: the answer is

_____. _____ signs: the answer is _____.



LEARN IT Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.

EXAMPLE 1

$3 \times (-2)$



→

YOUR TURN 1

$4 \times (-3)$



$(\quad) + (\quad) + (\quad) + (\quad)$

$= \quad$

EXPLAIN Why is $3 \times (-2)$ negative?

☐ A It is 3 groups of -2, which add to -6.

☐ B Multiplying always makes a number smaller.

EXAMPLE 2

-5×4

→

YOUR TURN 2

$6 \times (-7)$

different signs: _____

EXPLAIN Why does -5×4 give the same answer as $4 \times (-5)$?

☐ A The bigger number decides the sign.

☐ B Swapping the order does not change a product.

EXAMPLE 3

$-3 \times (-6)$

→

YOUR TURN 3

$-8 \times (-5)$

EXPLAIN $1 \times (-6) = -6$ and $0 \times (-6) = 0$: the answer went up by 6. What will $-1 \times (-6)$ be?

☐ A 6, because the answers keep going up by 6.

☐ B -6, because negatives give negatives.

CHECK Everyone answers. Circle one letter, then hold it up.

Work out $-4 \times (-5)$

(A) -20

(B) 20

(C) -9

(D) 9

**SPOT THE PATTERN** The first number goes down by 1 each time.

- 1 $3 \times (-4) =$ _____ 2 $2 \times (-4) =$ _____ 3 $1 \times (-4) =$ _____ 4 $0 \times (-4) =$ _____
- 5 $-1 \times (-4) =$ _____ 6 $-2 \times (-4) =$ _____ 7 $-3 \times (-4) =$ _____ 8 $-3 \times 4 =$ _____

What do you notice? In 1 to 7, the answer goes up by the same amount each time. What does that show about negative $\times$ negative?

**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 $4 \times (-3)$ _____ 2 -5×2 _____ 3 $-6 \times (-2)$ _____
- 4 $-7 + (-3)$ _____ 5 $7 \times (-4)$ _____ 6 $-3 \times (-9)$ _____
- 7 -8×5 _____ 8 $-7 \times (-3)$ _____ 9 $-6 - (-2)$ _____
- 10 $-10 \times (-4)$ _____ 11 $-2 \times 3 \times (-4)$ _____ 12 A diver goes down 5 m every minute for 6 minutes. What is her change in height? _____
- 13 $(-3) + (-3) + (-3) + (-3)$ _____ 14 $\square \times (-4) = 20$ _____ 15 Which is larger: -2×6 or $-2 \times (-6)$? _____

from 1.04

from 1.05

from 1.04

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.-40 -28 12 -12 -5 -30 m 40 21 27 -10 -10 $-2 \times (-6)$ -12 24 -4**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Find all the pairs of integers that multiply to -12 .

- 2 Always, sometimes or never? "Negative $\times$ negative $\times$ negative is negative." Explain.

- 3 Work out $(-1) \times (-1) \times (-1) \times (-1) \times (-1)$. What if there were 100 of them?

**EXIT TICKET** On your own. Hand it in.

- 1 -6×4 _____ 2 $-9 \times (-2)$ _____ 3 The temperature falls 3°C each hour for 5 hours. What is the total change?

Today I can multiply integers and choose the sign. After marking: _____ / 3

Dividing integers

Today I will divide integers and choose the sign.

NAME _____

CLASS _____

DATE _____

1.07

DO NOW From memory. No notes.

1 -6×7

2 $-4 \times (-8)$

3 $-2 - 9$

4 Order smallest to largest: $-4, 2, -8$

from 1.06

from 1.06

from 1.05

from 1.03

THINK FIRST Have a go.A debt of \$24 is shared equally by 4 people.
How much does each person owe?

Check it in Example 1.

WORDS

quotient: the answer to a division.

fact family: $3 \times (-4) = -12$, so
 $-12 \div 3 = -4$.

share: split into equal groups.

BIG IDEA Fill in the gaps with your teacher.Division _____ multiplication, so the sign rules are the _____. Same
signs: _____. Different signs: _____.**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

$-12 \div 3$

→

YOUR TURN 1

$-15 \div 5$

____ $\times$ (____) = ____

so ____ $\div$ ____ = ____

EXPLAIN How does knowing $3 \times (-4) = -12$ help you work out $-12 \div 3$?☐ A Division undoes multiplication: $-12 \div 3 = -4$.☐ B Divide 12 by 3 and leave out the sign.**EXAMPLE 2**

$28 \div (-4)$

→

YOUR TURN 2

$-36 \div 9$

different signs: _____

EXPLAIN How can you check that $28 \div (-4) = -7$?☐ A Divide without the signs: $28 \div 4 = 7$.☐ B Multiply back: $-7 \times (-4) = 28$.**EXAMPLE 3**

$-30 \div (-6)$

→

YOUR TURN 3

$-48 \div (-8)$

EXPLAIN Jo says $-30 \div (-6) = -5$. What has Jo done wrong?☐ A Same signs give a positive, so it is 5.☐ B Jo should have multiplied instead.**CHECK** Everyone answers. Circle one letter, then hold it up.Work out $-20 \div (-4)$ (A) -5 (B) 5 (C) -16 (D) 80

**SPOT THE PATTERN** Each question changes one sign or one number.

- 1 $12 \div 3 =$ _____ 2 $-12 \div 3 =$ _____ 3 $12 \div (-3) =$ _____ 4 $-12 \div (-3) =$ _____
- 5 $-12 \div (-4) =$ _____ 6 $-12 \div (-6) =$ _____ 7 $-12 \div 6 =$ _____ 8 $-120 \div (-6) =$ _____

What do you notice? Which answers are negative? What do their questions have in common?

**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 $-12 \div 3$ _____ 2 $18 \div (-6)$ _____ 3 $-24 \div (-8)$ _____
- 4 -9×3 _____ 5 $-35 \div 7$ _____ 6 $42 \div (-7)$ _____
- 7 $-36 \div (-4)$ _____ 8 $-64 \div (-8)$ _____ 9 $-4 - (-10)$ _____
- 10 $81 \div (-9)$ _____ 11 $-100 \div (-25)$ _____ 12 $\square \div 4 = -5$ _____
- 13 $-5 \times (-7)$ _____ 14 The temperature fell 18°C in 6 hours, the same each hour. What was the change each hour? _____
- 15 $-80 \div 4 \div (-2)$ _____

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.

9 -5 3 -4 -3°C -20 -9 8 -6 -27 -3 10 35 4 6

**GO FURTHER** Finished? Try these. Explain your thinking.

- 1 Find the average of -6 , 2 , -5 and -3 . (Add them, then divide by 4.)

- 2 Write three different divisions that have the answer -4 .

- 3 Always, sometimes or never? "Dividing makes a number smaller." Explain.

**EXIT TICKET** On your own. Hand it in.

- 1 $-35 \div 5$ _____ 2 $-72 \div (-8)$ _____ 3 A debt of \$60 is shared by 3 people. Write each share as an integer.

Today I can divide integers and choose the sign. After marking: _____ / 3

Order of operations with integers

Today I will use the order of operations with integers.

NAME _____

CLASS _____

DATE _____

1.08

DO NOW From memory. No notes.

1 $-18 \div 3$

from 1.07

2 $-4 \times (-6)$

from 1.06

3 $5 - 9$

from 1.05

4 $-8 + 3$

from 1.04

THINK FIRST Have a go.Two students work out $2 + 3 \times 4$. One gets 20.
One gets 14. Who is right?

Check it in Example 1.

WORDS**order of operations:**
brackets, then $\times$ and $\div$, then $+$ and $-$.**left to right:** for a tie, work from left to right.**evaluate:** work out the value.**BIG IDEA** Fill in the gaps with your teacher.**First,** work out the _____. **Next,** $\times$ and $\div$, from _____.**Last,** _____, from left to right. Use the integer rules at every _____.**LEARN IT** Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

$-3 + 4 \times 2$

→

YOUR TURN 1

$-10 + 3 \times 2$

$= ___ + ___$

$= ___$

EXPLAIN Why do we work out 4×2 before the adding?☐ A We always work from left to right.☐ B $\times$ and $\div$ come before $+$ and $-$.**EXAMPLE 2**

$(-3 + 7) \times 2$

→

YOUR TURN 2

$(1 - 6) \times 3$

$= ___ \times ___$

EXPLAIN Without brackets, $-3 + 7 \times 2 = 11$. Why do the brackets change the answer?☐ A Brackets go first, so we add before we multiply.☐ B Brackets always mean multiply.**EXAMPLE 3**

$6 - 2 \times (-3)$

→

YOUR TURN 3

$-5 - 4 \times (-2)$

EXPLAIN Mia says $6 - 2 \times (-3) = -12$. What has Mia done wrong?☐ A She should have added 6 and 2 first.☐ B She worked out $6 - 2$ first, but $\times$ comes first.**CHECK** Everyone answers. Circle one letter, then hold it up.Work out $10 - 3 \times (-2)$

(A) -14

(B) 4

(C) 16

(D) -16

**SPOT THE PATTERN** The same numbers, with one change each time.

1 $2 + 3 \times 4 =$ _____

2 $2 - 3 \times 4 =$ _____

3 $-2 - 3 \times 4 =$ _____

4 $-2 - 3 \times (-4) =$ _____

5 $(-2 - 3) \times (-4) =$ _____

6 $(-2 + 3) \times (-4) =$ _____

7 $-2 + 3 \times (-4) =$ _____

8 $-20 + 30 \times (-4) =$ _____

What do you notice? Find two questions with the same numbers but different answers. What made them different?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

1 $-4 + 2 \times 3$

2 $5 - 3 \times 4$

3 $-2 \times 3 + 1$

4 $-12 \div (-4)$

5 $(-2 + 5) \times 2$

6 $(1 - 4) \times 3$

7 $-12 \div 3 + 6$

from 1.07

8 $8 - 10 \div 2$

9 $-6 \times (-5)$

10 $-3 \times (-2) - 7$

11 $(-6 - 2) \div 4$

12 $6 - 2 \times (-5)$

from 1.06

13 $-3 - (-9)$

14 $-2 \times 3 + 4 \times (-2)$

15 $-3 \times (2 - 6)$

from 1.05

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.

2 6 -5 2 -14 16 -1 3 -9 3 -7 12 6 -2 30

**GO FURTHER** Finished? Try these. Explain your thinking.

1 Put brackets in to make it true: $-4 + 6 \times 3 = 6$

2 Use -2, 3 and 5 once each, with + and $\times$, and brackets if you like. What is the smallest answer you can make?3 Write a calculation with brackets, $\times$ and - that has the answer -12.**EXIT TICKET** On your own. Hand it in.

1 $-6 + 2 \times 4$

2 $(-6 + 2) \times 4$

3 $9 - 3 \times (-2)$

Today I can use the order of operations with integers. After marking: _____ / 3

Integers on a calculator

Today I will use a calculator with integers and check answers.

NAME _____

CLASS _____

DATE _____

1.09

DO NOW From memory. No notes.

1 $-3 + 2 \times 5$

from 1.08

2 $(4 - 9) \times 2$

from 1.08

3 $-56 \div 8$

from 1.07

4 $-7 \times (-6)$

from 1.06

THINK FIRST Have a go.Without a calculator: is -48×21 about -100 , about $-1\ 000$, or about $-10\ 000$?

Check it in Example 2.

WORDS**(-)** key: makes a number negative.**-** key: subtracts.**estimate**: a quick answer from rounded numbers.**BIG IDEA** Fill in the gaps with your teacher.

Use the _____ key to make a number negative, and the _____ key to subtract. _____ first. Then check the calculator's answer has the right _____ and size.

(-)

4

8

×

2

1

=

LEARN IT Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**

$-37 + 85$

→

YOUR TURN 1

$-126 + 59$

keys: **(-)** ____ + ____ =

= ____

EXPLAIN Why do we press **(-)** at the start, and not the **-** key?☐ A **(-)** makes 37 negative; **-** subtracts.☐ B Both keys do the same thing.**EXAMPLE 2**

-48×21

→

YOUR TURN 2

-39×31

estimate: ____ × ____ = ____

EXPLAIN How does the estimate show that $-1\ 008$ is a sensible answer?☐ A The estimate gives the exact answer.☐ B It is close to $-1\ 000$ and has the same sign.**EXAMPLE 3**

$(-35 - 47) \times 12$

→

YOUR TURN 3

$(-28 + 13) \times (-6)$

EXPLAIN Why must you type the brackets?☐ A Without them, it works out 47×12 first.☐ B The calculator cannot subtract without brackets.**CHECK** Everyone answers. Circle one letter, then hold it up.Ella presses **(-) 6 - 4 =** What does her calculator show?(A) -2 (B) 2 (C) -10 (D) 10

**SPOT THE PATTERN** Use your calculator. Each question changes one thing.

1 $-25 \times 4 =$ _____

2 $-25 \times 8 =$ _____

3 $-25 \times 12 =$ _____

4 $-25 \times 16 =$ _____

5 $-25 \times (-16) =$ _____

6 $25 \times (-16) =$ _____

7 $-250 \times 16 =$ _____

8 $-250 \times (-160) =$ _____

What do you notice? In 1 to 4, the second number goes up by 4 each time. What happens to the answer?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

1 $-347 + 128$

2 $-95 - 187$

3 -56×23

4 No calculator: $-6 + 3 \times (-2)$

5 $68 \times (-45)$

6 $-37 \times (-29)$

7 $-2\ 016 \div 42$

from 1.08

8 $-1\ 248 \div (-24)$

9 No calculator: $-45 \div (-9)$

10 $(-64 - 36) \times 15$

11 Estimate -62×19 . Then use your calculator.

12 A submarine at -185 m rises 12 m a minute for 9 minutes. Where is it now?

from 1.07

13 No calculator: $-8 \times (-7)$

14 Find the missing number:
 $-23 \times \square = 1\ 403$

15 $(-18 + 7) \times (-33)$

from 1.06

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.

-48 -3 060 -1 288 -219 -61 -77 m -1 500 52 1 073 -12 -282 363 56 -1 178 5

**GO FURTHER** Finished? Try these. Explain your thinking.

1 Use your calculator to find the missing number: $-37 \times \square = 1\ 147$

2 A student gets 1 008 for -48×21 . How can you tell it is wrong without working it out?

3 Moscow had these temperatures: -12 , -8 , -15 , -3 and -7°C . Find the average.

**EXIT TICKET** On your own. Hand it in.

1 $-234 + 97$

2 $-38 \times (-27)$

3 $(-45 - 27) \div 8$

Today I can use a calculator with integers and check answers. After marking: _____ / 3

Integer problems

Today I will solve problems with integers.

NAME _____

CLASS _____

DATE _____

1.10

DO NOW From memory. No notes.

1 $-144 \div 12$

2 $-2 \times (3 - 7)$

3 $-5 - (-8)$

4 Which is warmer: -14°C or -9°C ?

from 1.07

from 1.08

from 1.05

from 1.03

THINK FIRST Have a go.A lift at level -2 goes up 6 floors, then down 3 floors. Which level is it on now?

Check it in Example 1.

WORDS**balance:** how much money is in an account.**overdrawn:** the balance is below \$0.**change:** new – old.**BIG IDEA** Fill in the gaps with your teacher.

Read the story, then write a number _____. Up, rise, earn: _____.

Down, fall, spend: _____. The same change again and again: _____.

LEARN IT Copy your teacher's working for each example, then try your turn. Explain: tick the better reason.**EXAMPLE 1**It is -4°C at 6 am. By noon it is 9°C warmer. What is the temperature at noon?

→

YOUR TURN 1It is 3°C at 6 pm. By midnight it is 8°C colder. What is the temperature at midnight?

$$\begin{array}{r} ____ - ____ \\ = ____ ^{\circ}\text{C} \end{array}$$

EXPLAIN How did you know whether to add or subtract?☐ A Warmer is up, so add; colder is down, so subtract.☐ B A minus sign in the question means subtract.**EXAMPLE 2**

Sam has \$45. Sam spends \$70. What is the balance?

→

YOUR TURN 2Ana has $-\$30$. She is paid \$55. What is her balance?

$$____ + ____$$

EXPLAIN What does a balance of $-\$25$ mean for Sam?☐ A Sam owes the bank \$25.☐ B Sam has \$25 left.**EXAMPLE 3**A diver at -10 m goes down 3 m a minute for 4 minutes. Where is she now?

→

YOUR TURN 3A diver at -40 m rises 5 m a minute for 6 minutes. Where is she now?**EXPLAIN** Why do we multiply before we add?☐ A Multiplying always comes first in a story.☐ B The same change happens 4 times: $4 \times (-3)$.**CHECK** Everyone answers. Circle one letter, then hold it up.A gull is at 15 m. A shark is at -20 m . How far apart are they?☐ A 5 m☐ B 35 m☐ C -5 m ☐ D -35 m

**SPOT THE PATTERN** All four operations with -5 . Each changes one thing.

- 1 $-5 + 8 =$ _____ 2 $-5 + 5 =$ _____ 3 $-5 + 2 =$ _____ 4 $-5 - 2 =$ _____
- 5 $-5 - (-2) =$ _____ 6 $-5 \times 2 =$ _____ 7 $-5 \times (-2) =$ _____ 8 $-5 \div (-5) =$ _____

What do you notice? Which two questions have the same answer? Why?**MIXED PRACTICE** Mostly today's skill. Questions marked "from" are from earlier lessons. Show your working.

- 1 It is -5°C . It rises 8°C . What is the temperature now?
- 2 It is 4°C . It falls 9°C . What is the temperature now?
- 3 A lift is at level -3 . It goes up 5 floors. Which level is it on?
- 4 Order coldest first:
 -3°C , -13°C , 3°C , 0°C
- 5 You have \$10. You spend \$25. What is your balance?
- 6 Your balance is $-\$20$. You earn \$30. What is it now?
- 7 How far apart are 15 m and -10 m?
- 8 How far apart are -4 m and -19 m?
- 9 Which integer is halfway between -12 and 4?
- 10 A diver goes down 3 m a minute for 5 minutes. What is the change in depth?
- 11 It falls 2°C an hour for 6 hours. What is the total change?
- 12 A balance is \$20. You spend \$35, then earn \$10. What is the balance?
- 13 $-4 + 3 \times (-5)$
- 14 Quiz: +3 if right, -1 if wrong. Kai gets 6 right and 5 wrong. His score?
- 15 A plane is at 900 m. A submarine is at -150 m. How far apart are they?

from 1.03

from 1.02

from 1.08

CHECK YOUR ANSWERS All 15 answers, in a mixed-up order.25 m $-\$15$ level 2 3°C 13 $-\$5$ -15 m 15 m \$10 -13 , -3 , 0, 3 -5°C 1 050 m -19 -12°C -4 **GO FURTHER** Finished? Try these. Explain your thinking.

- 1 In the quiz, Lea answered 10 questions and scored 14. How many were right?
- 2 It is -6°C . It warms 2°C an hour for 12 hours, then cools 3°C an hour for 4 hours. Temperature?

**EXIT TICKET** On your own. Hand it in.

- 1 It is -7°C . It rises 11°C . What is the temperature now?
- 2 A balance is \$15. You spend \$40. What is the balance?
- 3 A diver at -8 m goes down 2 m a minute for 5 minutes. Where is she?

Today I can solve problems with integers. After marking: _____ / 3

Chapter review

Today I will show what I remember from the whole chapter.

NAME _____

CLASS _____

DATE _____

Review

MIXED REVIEW A No notes. Show your working. Calculator only where it says so.

1 What is the opposite of -17 ?

1.01

3 How far is it from -6 to 9 ?

1.02

5 Write $<$ or $>$: -12 $\square$ -21

1.03

7 $-9 + 14$

1.04

9 $-4 - 7$

1.05

11 -7×8

1.06

2 $-(-8)$

1.01

4 Which integer is halfway between -14 and 2 ?

1.02

6 Order smallest to largest: $5, -8, 0, -3, -10$

1.03

8 $3 + (-11)$

1.04

10 $-2 - (-10)$

1.05

12 $-6 \times (-9)$

1.06

**MIXED REVIEW B** Keep going. Each question shows which lesson it comes from.

13 $-81 \div 9$

1.07

15 $-5 + 3 \times (-4)$

1.08

17 Calculator: $-384 \div 16$

1.09

19 It is -3°C . It falls 9°C . What is the temperature now?

1.10

21 How far apart are 40 m and -35 m?

1.10

23 Is $6 - (-4)$ the same as $6 + 4$? Explain.

1.05

14 $-64 \div (-8)$

1.07

16 $(-5 + 3) \times (-4)$

1.08

18 Calculator: $(-73 + 28) \times 11$

1.09

20 A diver at -25 m rises 4 m a minute for 5 minutes. Where is she now?

1.10

22 $-2 \times (-3) \times (-5)$

1.06

24 Put brackets in to make it true:
 $2 - 8 \times 3 = -18$

1.08

**WHAT NEXT?** Mark your answers with your teacher. Tick each lesson where you got every question right.

1.01

☐

1.02

☐

1.03

☐

1.04

☐

1.05

☐

1.06

☐

1.07

☐

1.08

☐

1.09

☐

1.10

☐

Not ticked? Go back to that lesson's examples, then try its exit ticket again.